

P3↔P2 Mating Primitive Inheritance

Series: CC.13 — Paper 3 ↔ Paper 2 Cross-Derivation Note (#679)

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

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This work derives from and formalizes commitments across the CKS theory trilogy: *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026), and *Inter-Self Coordination via Shared Substrate / Full Aspect Integration* (Li, April 2026).

Abstract

Paper 3's Full Aspect Integration (FAI) is not a new coordination primitive. It is Paper 2's mating event extended to inter-Self scope. This note formalizes that inheritance relationship across three axes. First, Paper 3's three merge patterns (D1.08–D1.09) map directly and one-to-one onto Paper 2's three mating variants (B1.06): full merger inherits full union; selective merge inherits selective merge; provenance-carry-over merge inherits lineage-preserved union. The three-variant structure was inherited from Paper 2; only the naming adapted to the inter-Self scope. Second, the evolution feed (D1.17–D1.21) is the inter-Self scope adaptation of the mating event's offspring production: where intra-Self mating produces new governed content within one Self's governance architecture, FAI produces learning outputs that return to each participating Self's home substrate. The evolution feed achieves at inter-Self scope what offspring achieves at intra-Self scope. Third, the three-tier conflict handling mechanism (D0.03) extends Paper 2's intra-Self mating conflict resolution to the inter-Self boundary, where conflicts between different Selves' governance content require the more structured preserve–resolve–escalate framework. Establishing FAI as the inter-Self mating primitive is the foundational inheritance claim: it makes the entire Paper 3 architecture follow from Paper 2's mating event, rather than introducing a new coordination primitive that would require fresh defense.

1. Inheritance Identification

Source: Paper 2, §6.3 (B1.06) — the mating event as the intra-Self coordination primitive producing new governance content through the temporary interaction of two or more aspects within one Self.

Extension: Paper 3, §5 (D1.12) — Full Aspect Integration (FAI) as the inter-Self mating analog; Paper 3, §5 (D1.08–D1.09) — the three merge patterns.

The inheritance claim is straightforward: FAI is mating at inter-Self scope. Paper 2 established mating as the governed lifecycle primitive over content combination — two aspects interacting in a temporary space, with their interaction producing new governance outputs, configured by orchestration substrate, available in three pattern variants. Paper 3 extends that primitive to the inter-Self perimeter. When two aspects from different Selves interact through the shared substrate during an FAI event, the architectural posture is identical to Paper 2's mating event. The scope is different; the primitive is the same.

This inheritance identification is defensively important. An adversary could claim that FAI is a novel coordination primitive unrelated to Paper 2's mating operation, which would require fresh defense of FAI's architectural commitments. The inheritance forecloses that claim: FAI is the Paper 2 merge primitive extended to inter-Self scope, and every architectural commitment that Paper 2 established for mating carries through to FAI without redefense at the mating-primitive level.

2. Source Commitment at Paper 2 Scope

Paper 2 §6.3 establishes mating as one governable primitive over content combination (B1.06). The mating event is the second of three lifecycle primitives — birth, mating, death — that Paper 2 names as governed operations at every level of the CKS architecture (cell, aspect, Self). As a lifecycle primitive, mating is not application-level merge logic bolted onto version control infrastructure; it is a first-class architectural operation with its own governance properties, its own conflict-handling commitments, and its own three-variant structure.

The three mating variants Paper 2 establishes are:

Full union. Both aspects contribute all their content; the resulting state merges both. Merge-time conflicts become persistent first-class substrate state in the offspring, inheriting Paper 1's conflict-as-first-class commitment. The union pattern does not require conflict resolution to complete the merge; conflicts carry forward as substrate content the offspring holds.

Selective merge. Aspects contribute selectively; specific content is chosen for inclusion by humans or LLMs operating under human-authored governance rules. The curation rules are themselves substrate content under human authority — this is governance-driven curation, not constraint-driven composition.

Lineage-preserved union. Content from both aspects is retained with cross-references preserving each lineage. The offspring carries pointers to parent cells so every element of its substrate has traceable derivation back to its source. Lineage pointers are substrate content, not a separate provenance graph layered above it.

The governance properties of mating at Paper 2 scope hold across all three variants: the combination is configured by orchestration substrate; conflict handling is part of the interaction; the output — offspring — is new governed content within the same Self's governance architecture. The intra-Self scope is the structural constraint: both aspects belong to one Self, and the offspring they

produce lives within that Self's governance perimeter.

3. Extension to Inter-Self Scope

Paper 3's FAI extends mating to the inter-Self perimeter along three dimensions.

Three merge patterns inherit three mating variants

The mapping is direct and one-to-one:

Full merger (D1.08) ← Full union (B1.06): aspects from different Selves contribute all their content to the shared substrate; the architectural default in Paper 3 is full merge.

Selective merge (D1.09, Pattern 2) ← Selective merge (B1.06): governance-configured partial merge, with each Self's governance specifying which aspects and which content to contribute.

Provenance-carry-over merge (D1.09, Pattern 3) ← Lineage-preserved union (B1.06): explicit provenance carry-over at the inter-Self perimeter, with provenance depth governance-configured by the contributing Self.

The three-variant structure is inherited from Paper 2's mating event. The naming adapted to the inter-Self scope — "full merger" reflects the architectural default Paper 3 names that Paper 2 did not name among its variants; "provenance-carry-over" reflects the inter-Self provenance context — but the governance properties of each variant are identical to their Paper 2 analogs. An adversary cannot claim that Paper 3's three merge patterns are novel architectural commitments, because they are Paper 2's three mating variants renamed at inter-Self scope.

Evolution feed replaces offspring as the output

This is the key adaptation that inter-Self scope requires. At intra-Self scope, mating produces offspring: new governed content within the same Self's governance architecture. The offspring lives in one governance space; it is a new cell, aspect, or Self-level entity with its own substrate. At inter-Self scope, FAI cannot produce offspring in the same sense, because the "offspring" would need to be in each participating Self's home substrate — not in a new unified governance space that does not exist.

The evolution feed (D1.17–D1.21) is what solves this. Rather than producing new content within one governance space, FAI produces learning outputs that return to each participating Self's home substrate through Paper 2's evolution mechanisms. Action-feedback evolution from FAI: outcomes recorded during an FAI event become new action-layer content in each participating Self's home substrate, processed by that Self's action-feedback evolution mechanism under its home governance. DNA evolution from FAI: where governance authorizes it, orchestration patterns, schemas, or rules from another Self's contributed aspects can be selectively absorbed into the receiving Self's DNA through directed selection under home governance. The FAI event is, from each Self's perspective, a class of lived experience that produced action-layer records and potentially DNA-layer candidates.

The evolution feed achieves at inter-Self scope what offspring achieves at intra-Self scope: it is the mechanism through which the mating event's output returns to and becomes part of the governance architecture of the entities that participated. The adaptation is required by the scope; the function is inherited from the mating event's offspring production.

Conflict handling elevates to the three-tier mechanism

Paper 2's mating event had governance mechanisms for handling content conflicts within one Self — the intra-Self conflict handling framework that Paper 2 extends from Paper 1's conflict-as-first-class commitment. At intra-Self scope, merge-time conflicts become substrate content the offspring carries, with cell-level orchestration rules in the offspring governing how they are handled.

At inter-Self scope, conflicts between different Selves' governance content require a more structured framework, because the conflicts may involve content governed by separate authority structures with no shared orchestration rules for the conflict class. Paper 3's three-tier mechanism (D0.03) — preserve, resolve via orchestration, escalate to humans — is the inter-Self scope extension of Paper 2's intra-Self mating conflict resolution. The preserve tier inherits Paper 1's conflict-as-first-class commitment directly. The resolve-via-orchestration tier inherits Paper 2's cell-level conflict resolution through the shared substrate's orchestration rules. The escalate-to-humans tier inherits Paper 1's human-governance commitment at the joint-authority level, reaching across both Selves' governance authorities for conflicts that no prebuilt orchestration logic resolves. The three-tier mechanism is not a new conflict resolution architecture; it is what Paper 2's mating conflict resolution produces when extended to the inter-Self boundary.

4. What Is Preserved and What Is Extended

Preserved from Paper 2's mating event:

The mating event as the coordination primitive — two or more aspects interacting in a temporary space to produce coordination outputs.

Three pattern variants (now: three merge patterns) with identical governance properties.

Conflict handling as part of the interaction, inheriting Paper 1's conflict-as-first-class commitment.

Exchange bounded to substrate content: DNA-layer and action-layer content; instinct-layer content and LLM weights do not cross the inter-Self perimeter, per Paper 2's instinct/reasoning separation.

Orchestration-substrate configurability: every dimension of the FAI protocol is governance-configured under human authority.

Extended at inter-Self scope:

The scope: aspects from different Selves, not aspects within one Self.

The output: evolution feed to multiple home substrates, not offspring in one governance space.

The conflict handling complexity: three-tier mechanism across joint authority, not intra-Self resolution under one Self's orchestration rules.

The architectural default: Paper 3 names full merge as the default among the three pattern variants; Paper 2 did not name a default at intra-Self scope. This is the one fresh commitment at the merge-primitive level that Paper 3 adds.

5. Prior-Art Inheritance Claim

FAI as the inter-Self mating primitive directly inherits from Paper 2's B1.06 mating event. The three merge patterns inherit the three mating variants. The evolution feed inherits the offspring-production function of the mating event, adapted to the inter-Self scope where outputs must re-enter multiple existing governance spaces rather than populating one new one. The three-tier conflict handling inherits and elevates Paper 2's intra-Self mating conflict resolution to the inter-Self boundary.

The foundational claim is: FAI is not a new coordination primitive. It is the mating primitive at the next scope level. Establishing this inheritance is what makes the entire Paper 3 architecture follow from Paper 2's mating event — every architectural commitment FAI carries is either inherited from Paper 2's mating event or is a named extension at inter-Self scope. The no-new-axioms discipline at the mating-primitive level holds: Paper 3 extends the scope of the primitive; it does not introduce a new one.

Source papers: Li, Wenxin. "The Instinct/Reasoning Separation Outside the Model." April 2026. Li, Wenxin.

"Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems." April 2026. Li, Wenxin.

"Inter-Self Coordination via Shared Substrate / Full Aspect Integration." April 2026.